Detailed Test Design and Execution

1. Selected Major Module

This document treats `coupon_discount_engine` as the selected major feature of the target e-commerce promotion application. It is not the whole AutoTestDesign tool itself. Instead, it is the executable feature used to demonstrate that the test designs produced or refined with ARG-Test can be translated into concrete black-box and white-box tests.

This module is a strong detailed-execution target because it combines:

- valid and invalid input partitions
- multiple numeric boundaries
- interacting business rules
- explicit expected results
- a compact reference implementation suitable for white-box execution

The module is suitable for detailed execution because its expected behavior is explicit, reviewable, and rich enough to require more than one testing technique. A single nominal-path test would not be sufficient: the module has rejection rules, threshold rules, membership restrictions, and implementation branches that must be checked together.

2. Requirement Basis and Scope

2.1 Formalized Requirement Summary

The detailed execution document uses the following normalized rules derived from the final requirement set and the selected reference implementation:

- R1: only one coupon may be applied to an order
- R2: unknown coupon codes must be rejected
- R3: expired coupons must be rejected
- R4: `SAVE10` requires `subtotal >= 50`
- R5: `SAVE20` requires `subtotal >= 100`
- R6: `SAVE20` also requires premium membership
- R7: `SAVE20` cannot be combined with sale items
- R8: `FREESHIP` requires `subtotal >= 30`
- R9: no-coupon input should leave subtotal and shipping unchanged

These rules define the coverage items used in the rest of the document. The black-box design focuses on externally visible behavior from these rules, while the white-box design checks whether the executable reference implementation exercises all important branches that implement them.

2.2 Implementation Under Test

Item	Path
Reference implementation	`reference_impl/coupon_discount_engine.py`
Black-box tests	`tests/test_coupon_discount_engine_blackbox.py`
White-box tests	`tests/test_coupon_discount_engine_whitebox.py`
Seeded mutants	`reference_impl/coupon_discount_engine_mutants.py`

3. Test Environment and Tooling

Tool	Purpose
`pytest`	execute black-box and white-box test cases
`coverage.py`	collect statement and branch coverage
mutant functions	demonstrate defect detection usefulness

`pytest` was selected because the reference implementation is a compact Python module and the expected outcomes can be expressed directly with assertions. `coverage.py` was added to connect the test design with executable evidence. Branch coverage is especially useful here because coupon logic is implemented through conditionals, not only straight-line statements.

Commands used:

```
python -m pytest tests\test_coupon_discount_engine_blackbox.py tests\test_coupon_discount_engine_whitebox.
python -m pytest tests -q
python -m coverage run --branch -m pytest tests -q
python -m coverage report -m reference_impl\coupon_discount_engine.py
python -m coverage xml -o final_docs\execution_evidence\coupon_discount_engine_coverage.xml
python -m coverage xml -o final_docs\execution_evidence\coupon_discount_engine_branch_coverage.xml
```

4. Black-Box Test Design

The black-box design was derived from the requirement rules without relying on internal implementation details. Three techniques are used together because they cover different risks. Equivalence Partitioning separates valid and invalid classes. Boundary Value Analysis targets monetary thresholds where off-by-one or inclusive/exclusive mistakes are likely. Decision Table Testing captures business-rule combinations, especially for `SAVE20`, where multiple conditions interact.

4.1 Equivalence Partitioning

Partition ID	Partition type	Representative condition	Designed test(s)	Expected result
EP1	valid	no coupon provided	`BB01`	accept and keep subtotal/shipping unchanged
EP2	invalid	more than one coupon provided	`BB02`	reject with one-coupon-only reason

Partition ID	Partition type	Representative condition	Designed test(s)	Expected result
EP3	invalid	unknown coupon code	`BB03`	reject with unknown-coupon reason
EP4	invalid	expired coupon	`BB04`	reject with expired-coupon reason
EP5	invalid	premium-only coupon used by non-premium customer	`BB07`	reject with membership reason
EP6	invalid	restricted coupon combined with sale items	`BB08`	reject with sale-item restriction reason
EP7	valid	`SAVE20` with all preconditions satisfied	`BB09`	accept and apply 20% discount

4.2 Boundary Value Analysis

Boundary ID	Threshold	Below	On boundary	Above / valid representative	Designed test(s)
B1	`SAVE10` subtotal `50`	`49`	`50`	`60`	`BB05`, `BB06`, `WB05`
B2	`SAVE20` subtotal `100`	`99`	`100`	`120`	`WB03`, `BB09`
B3	`FREESHIP` subtotal `30`	`29`	`30`	`80` with no coupon	`WB04`, `BB10`, `BB01`

4.3 Decision Table

Rule	Coupon	Threshold met	Premium member	Sale items present	Expired	Expected outcome
D1	none	N/A	N/A	N/A	N/A	accept, no change
D2	`SAVE10`	no	N/A	N/A	no	reject
D3	`SAVE10`	yes	N/A	N/A	no	accept, 10% discount
D4	`SAVE20`	yes	no	no	no	reject
D5	`SAVE20`	yes	yes	yes	no	reject
D6	`SAVE20`	yes	yes	no	no	accept, 20% discount
D7	`FREESHIP`	no	N/A	N/A	no	reject
D8	`FREESHIP`	yes	N/A	N/A	no	accept, shipping becomes zero

4.4 Executable Black-Box Cases

Test ID	`pytest` function	Main technique	Covered rule / purpose
BB01	`test_no_coupon_keeps_order_values`	EP	no-coupon valid partition
BB02	`test_multiple_coupons_are_rejected`	EP	one-coupon-only rejection

Test ID	`pytest` function	Main technique	Covered rule / purpose
BB03	`test_unknown_coupon_is_rejected`	EP	unknown coupon rejection
BB04	`test_expired_coupon_is_rejected`	EP	expired coupon rejection
BB05	`test_save10_boundary_below_threshold_is_rejected`	BVA	`SAVE10` just below threshold
BB06	`test_save10_boundary_on_threshold_is_accepted`	BVA	`SAVE10` exact threshold acceptance
BB07	`test_save20_requires_premium_membership`	EP / decision table	premium requirement
BB08	`test_save20_with_sale_items_is_rejected`	EP / decision table	sale-item restriction
BB09	`test_save20_valid_case_applies_discount`	decision table	valid `SAVE20` acceptance
BB10	`test_freeship_threshold_on_boundary_sets_shipping_to_zero`	BVA	`FREESHIP` exact threshold acceptance

4.5 Requirement-to-Test Traceability

The following traceability matrix connects the requirement rules, coverage ideas, selected techniques, executable test IDs, and expected behavior. This is the main bridge between the abstract test design and the concrete `pytest` implementation.

Requirement	Coverage item	Technique	Test ID(s)	`pytest` function(s)	Expected behavior
R1	more than one coupon	EP	`BB02`	`test_multiple_coupons_are_rejected`	reject multiple coupons
R2	unknown coupon code	EP	`BB03`	`test_unknown_coupon_is_rejected`	reject unknown code
R3	expired coupon	EP	`BB04`	`test_expired_coupon_is_rejected`	reject expired coupon
R4	`SAVE10` below/on threshold	BVA	`BB05`, `BB06`, `WB05`	`test_save10_boundary_below_threshold_is_rejected`, `test_save10_boundary_on_threshold_is_accepted`, `test_coupon_normalization_accepts_mixed_case_and_spacing`	reject below 50, accept at 50 or above
R5	`SAVE20` threshold	BVA / decision table	`WB03`, `BB09`	`test_save20_below_threshold_keeps_original_values`, `test_save20_valid_case_applies_discount`	reject below 100, accept when all conditions hold

Requirement	Coverage item	Technique	Test ID(s)	`pytest` function(s)	Expected behavior
R6	premium membership required	EP / decision table	`BB07`	`test_save20_requires_premium_membership`	reject non-premium customer
R7	sale-item restriction	EP / decision table	`BB08`	`test_save20_with_sale_items_is_rejected`	reject `SAVE20` with sale items
R8	`FREESHIP` threshold	BVA	`WB04`, `BB10`	`test_freeship_below_threshold_is_rejected`, `test_freeship_threshold_on_boundary_sets_shipping_to_zero`	reject below 30, set shipping to zero at 30
R9	no coupon path	EP	`BB01`	`test_no_coupon_keeps_order_values`	accept and keep values unchanged

5. White-Box Test Design

5.1 White-Box Objectives

The white-box design complements the black-box suite by checking implementation branches that may not be fully justified by a single external rule. For example, the negative value guards are implementation-level safety checks, while coupon normalization verifies that the function handles user-facing input variations such as whitespace and mixed letter case.

The white-box design targets:

- input guard branches for negative values
- all coupon dispatch branches
- rejection branches for each invalid condition
- acceptance branches for `SAVE10`, `SAVE20`, and `FREESHIP`
- normalization behavior for mixed-case and whitespace coupon input

5.2 Branch-to-Test Mapping

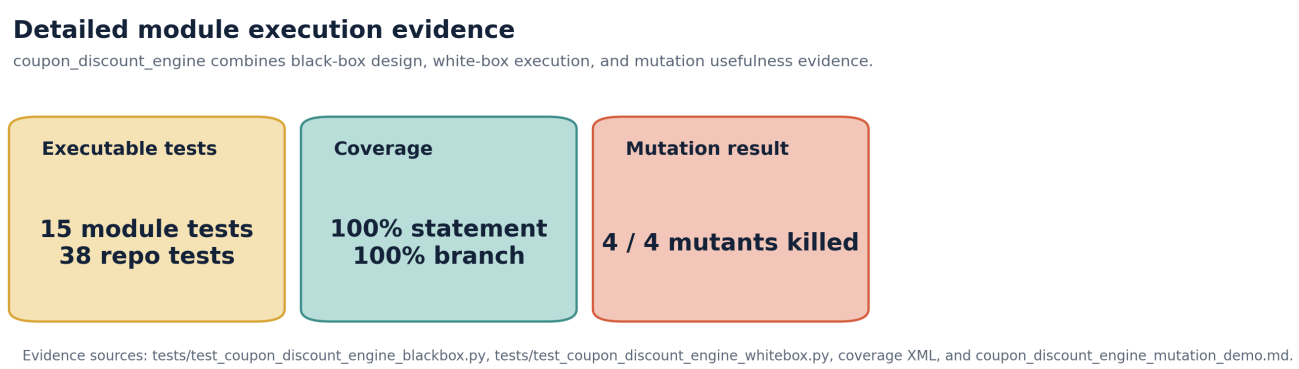
White-box obligation	Designed test(s)
`subtotal < 0` guard raises `ValueError`	`WB01`
`shipping_fee < 0` guard raises `ValueError`	`WB02`
`SAVE20` threshold-reject branch	`WB03`
`FREESHIP` threshold-reject branch	`WB04`
coupon normalization branch	`WB05`
`SAVE10` accept path	`BB06`, `WB05`
`SAVE20` accept path	`BB09`
`FREESHIP` accept path	`BB10`

5.3 Executable White-Box Cases

Test ID	`pytest` function	Covered white-box purpose
WB01	`test_negative_subtotal_raises_value_error`	negative subtotal guard
WB02	`test_negative_shipping_fee_raises_value_error`	negative shipping fee guard
WB03	`test_save20_below_threshold_keeps_original_values`	`SAVE20` threshold rejection branch
WB04	`test_freeship_below_threshold_is_rejected`	`FREESHIP` rejection branch
WB05	`test_coupon_normalization_accepts_mixed_case_and_spacing`	normalization and `SAVE10` valid path

Together, the black-box and white-box tests form a layered design. The black-box part proves that the observable business rules are represented. The white-box part proves that the implementation paths behind those rules are executed, including defensive branches and normalization behavior.

6. Execution Results



Detailed Module Execution Evidence

6.1 Summary of Observed Results

Item	Observed result
Module-specific executable tests	`15 passed`
Repository regression suite at report-preparation time	`38 passed`
Statement coverage on the reference module	`100%`
Branch coverage on the reference module	`100%`
Mutation result	`4 / 4 mutants killed`

6.2 Coverage Summary

```
Name                                    Stmts   Miss Branch BrPart  Cover
reference_impl\coupon_discount_engine.py      51      0     26      0   100%
TOTAL                                         51      0     26      0   100%
```

Coverage interpretation:

- black-box design is strong enough to exercise the functional rule structure
- white-box design closes the remaining branch obligations
- no branch in the selected reference implementation remains unexecuted

6.3 Result Analysis

The detailed execution result is strong for three reasons.

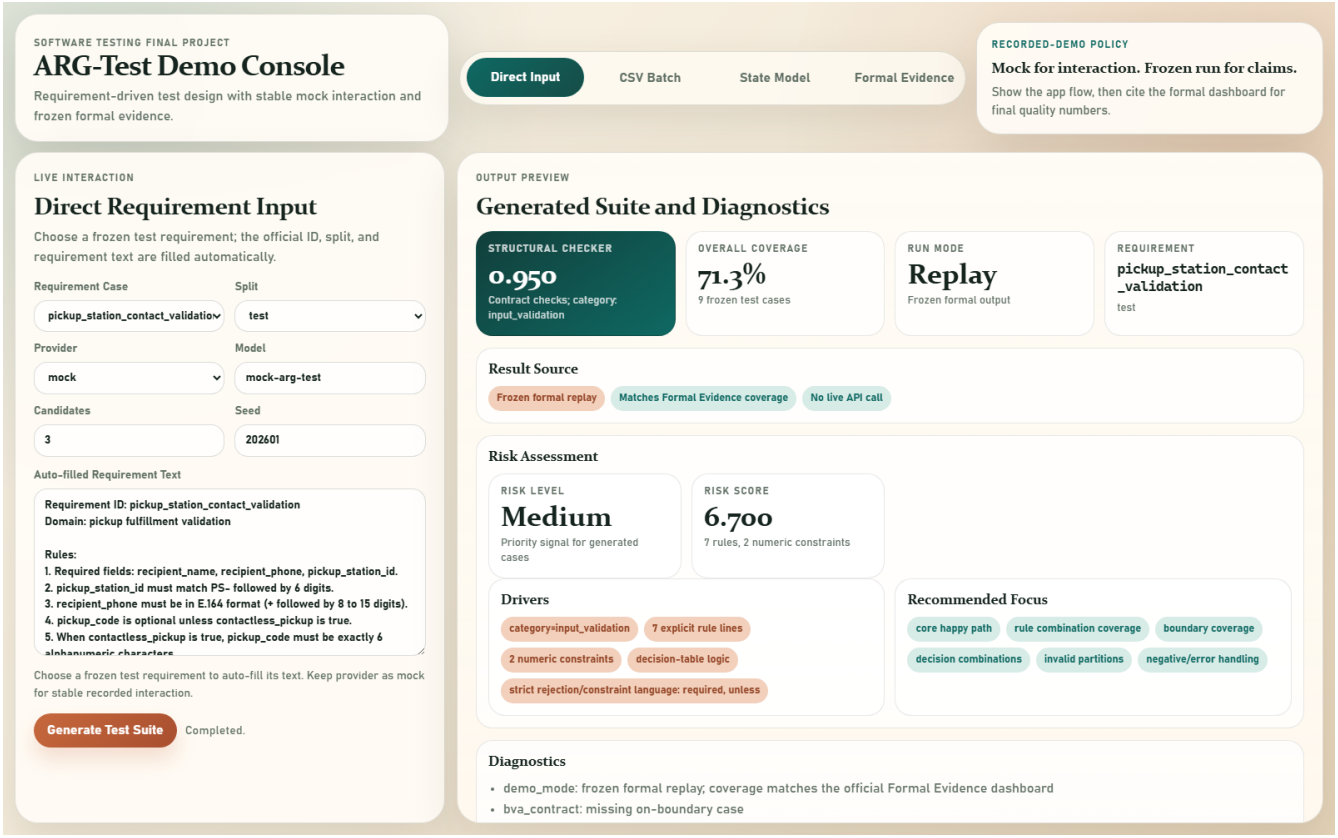
First, the black-box suite is not superficial. It covers valid partitions, invalid partitions, multiple thresholds, and interacting rule conditions such as premium membership plus sale-item restrictions.

Second, the white-box suite is not decorative. It exercises the explicit negative-input guards and rejection/acceptance branches that are easy to overlook in a purely requirement-level discussion.

Third, the combined suite is compact. With only 15 module-focused cases, it achieves complete statement and branch coverage on the selected implementation, which is a good tradeoff between completeness and maintainability for this module-level validation.

7. Web Demo Consistency Validation

The final delivery also validates that the Web demo presents the same formal evidence used by the report instead of mixing official results with unrelated local mock outputs. This check matters because the demo is the primary presentation surface for the final defense, while the formal result bundle remains the source of quantitative evidence.

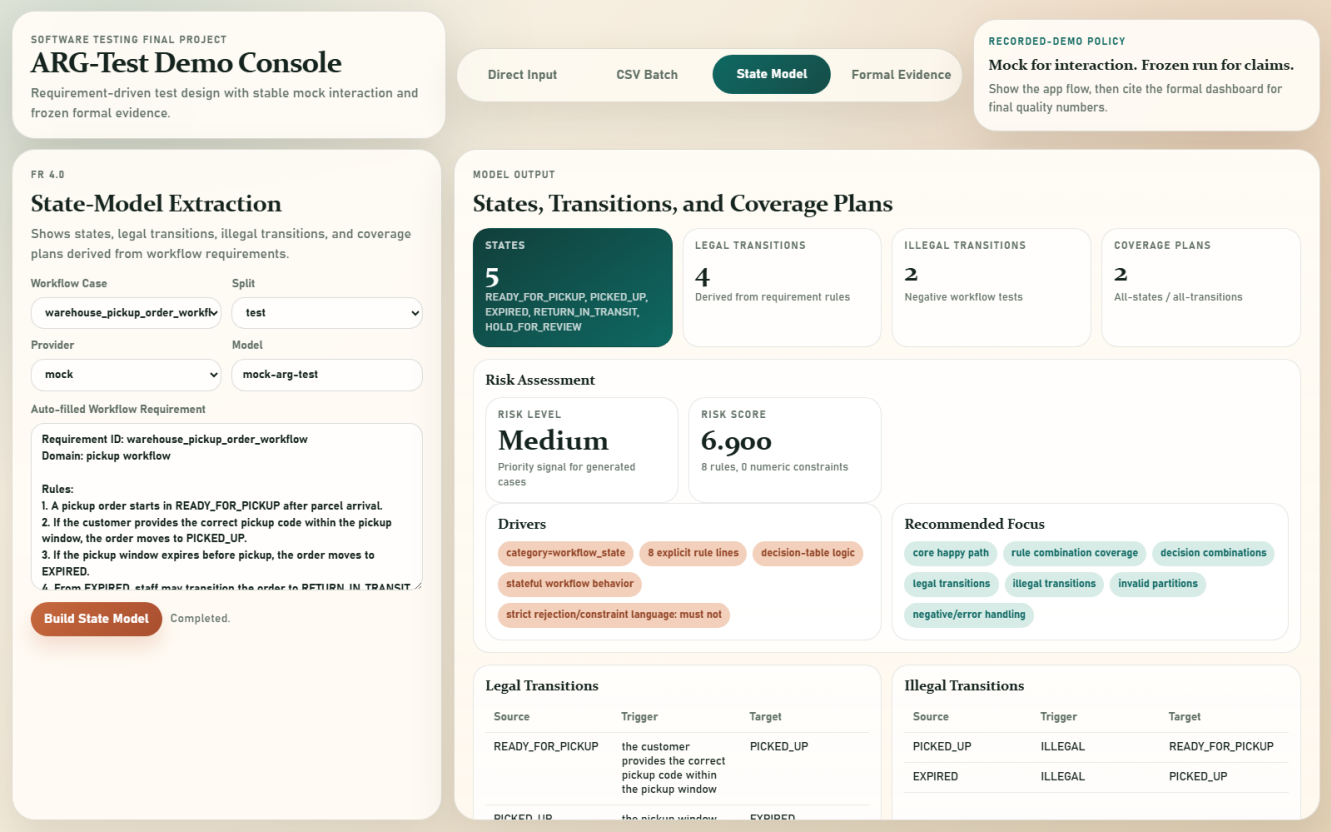


Direct Frozen Replay Evidence

The Direct tab now uses a formal requirement catalog. When a built-in formal test requirement is selected, the backend returns `frozen_formal_run` replay data from the archived result snapshot. This prevents mismatches such as a selected label showing one coverage value while the generated local mock result displays another value.

Web demo consistency check	Observed result
Direct-tab formal catalog replay	`16 / 16` formal test requirements replayed frozen outputs
CSV sample replay	`2 / 2` matching CSV rows replayed frozen outputs
Formal dashboard aggregate	Average overall coverage shown as `61.5%`
Repository regression after Web demo fixes	`38 passed`

The State-Model tab was also checked on the five workflow examples included in the demo catalog. Each catalog workflow now produces a non-empty legal-transition model, and illegal transitions are only reported when the requirement text explicitly supports them.



State Model Extraction Evidence

Workflow example	States	Legal transitions	Illegal transitions
`order_approval_state_machine`	6	7	0
`order_split_shipment_state_machine`	4	3	2
`payment_3ds_authentication_flow`	6	6	2
`return_exchange_approval_workflow`	9	8	1
`warehouse_pickup_order_workflow`	5	4	2

This Web-level validation does not replace the detailed executable module evidence. Instead, it protects the final demonstration from interpretation errors: official quantitative results come from frozen formal replay, while ad hoc edited inputs remain clearly labeled as local mock generation.

8. Mutation-Based Usefulness Demonstration

The evaluation also includes defect-seeded usefulness evidence. Four representative mutants were created:

- a mutant that allows multiple coupons
- a mutant that rejects `SAVE10` exactly at subtotal `50`
- a mutant that ignores the `SAVE20` sale-item restriction
- a mutant that rejects `FREESHIP` exactly at subtotal `30`

Observed outcome:

- `4 / 4` mutants were killed

Mutation-to-test mapping:

Mutant	Killed by
`multiple_coupons_allowed`	`BB02`, `BB08`
`save10_boundary_bug`	`BB06`
`save20_sale_item_bug`	`BB08`
`freeship_boundary_bug`	`BB10`

This is important because it shows that the detailed suite is not only structurally complete. It is also effective at detecting realistic logic defects that correspond to the selected business rules and boundaries.

9. Evidence Paths

Primary evidence files:

- `final_docs/execution_evidence/coupon_discount_engine_execution_summary.md`
- `final_docs/execution_evidence/coupon_discount_engine_coverage.xml`
- `final_docs/execution_evidence/coupon_discount_engine_branch_coverage.xml`
- `final_docs/execution_evidence/coupon_discount_engine_mutation_demo.md`
- `demo_web/app.py`
- `tests/test_demo_web_api.py`
- `report_assets/final_demo_package/frontend_focus/formal_results_snapshot/`

10. Conclusion

The `coupon_discount_engine` module is supported by a complete detailed-design and execution chain. The module is covered by multiple black-box techniques, executable white-box tests, complete statement and branch coverage, and a successful mutation demonstration. This makes it a credible detailed anchor for the overall ARG-Test submission rather than a merely illustrative example.

Most importantly, the evidence chain is complete: requirement rules are transformed into coverage items, coverage items are mapped to executable `pytest` tests, the tests are run against a reference implementation, and the resulting suite is checked with both coverage and mutation evidence. The result is a traceable module-level validation package that connects test case design, test tool implementation, and test result analysis.